

Royce Nguyen

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EDUCATION

University of California, San Diego

La Jolla, CA

Bachelor of Science in Data Science | GPA: 3.84/4

Expected June 2027

Relevant Coursework: Probabilistic Modeling and ML, Relational Data Management, ML: Learning Algorithms

EXPERIENCE

Data Science Researcher

January 2026 – April 2026

UC San Diego Natural Reserve System

San Diego, CA

- Built a scoring model ranking 2,700+ San Diego locations by undiscovered species potential to advise observer deployment.
- Identified under observed habitat clusters across 10 global cities, informing prioritization strategy for San Diego deployments.
- Deployed an interactive web app used by UCSD Natural Reserve System staff to explore priority zones and plan volunteer strategy.

Data Analyst Intern

June 2025 – August 2025

ECGO

Atlanta, GA

- Queried 30,000+ recycling submissions using SQL/PostgreSQL to identify key user behavior segments and patterns that informed retention and engagement strategy.
- Deployed a pretrained Roboflow CV model for brand name detection on noisy, multi object recycling submissions, achieving 70% accuracy and cutting manual review time.
- Partnered with UX, Sales, Marketing, and Analytics teams to translate data findings into actionable experiments, driving product and campaign improvements.

Data Scientist

February 2024 – February 2025

Biokind Analytics

San Diego, CA

- Analyzed 20,000+ donor records for nonprofit leaders, identifying actionable patterns to guide outreach and fundraising decisions.
- Built a LangChain workflow using the OpenAI API and GPT-4o to generate personalized donor letters, reducing drafting time by 50%.
- Developed an automated data cleaning pipeline and live web dashboard delivering real-time user metrics to executives.

PROJECTS

Scikit-Learn: Under the Hood | Python, FastAPI, React, D3.js

October 2025 – December 2025

- Deployed a full-stack FastAPI and React learning app on Render that visualizes ML algorithms with step-by-step execution traces, helping students understand model behavior beyond black-box outputs.
- Built interactive D3.js/Plotly visualizations that respond in real-time to parameter changes, breaking down complex algorithm formulas.
- Implemented RESTful API endpoints to serve algorithm state and computation steps as JSON for frontend.
- Used by 200+ students in DSC 140A (Probabilistic Modeling and Machine Learning) office hours as a hands on reference for understanding algorithm behavior.

League of Legends Tower Analysis | Python, Scikit-learn, Matplotlib, Pandas

March 2025 – April 2025

- Engineered predictive features from 150k+ League of Legends match records, using EDA and hypothesis testing to identify early-game indicators that correlate with late-game tower advantages.
- Developed an end-to-end machine learning pipeline to predict a team's end-of-match tower count, achieving an R^2 of 0.86.
- Conducted fairness analysis using permutation tests to verify consistent model performance across a diverse set of teams.

TECHNICAL SKILLS

Languages: Python, Java, C#, R, SQL, JavaScript, HTML, CSS, TypeScript

Developer Tools: AWS, Git, Jupyter Notebook, Microsoft Excel, Tableau

Frameworks/Libraries: Django, Pandas, NumPy, Plotly, Scikit-learn, Seaborn, BeautifulSoup4, PyTorch, React